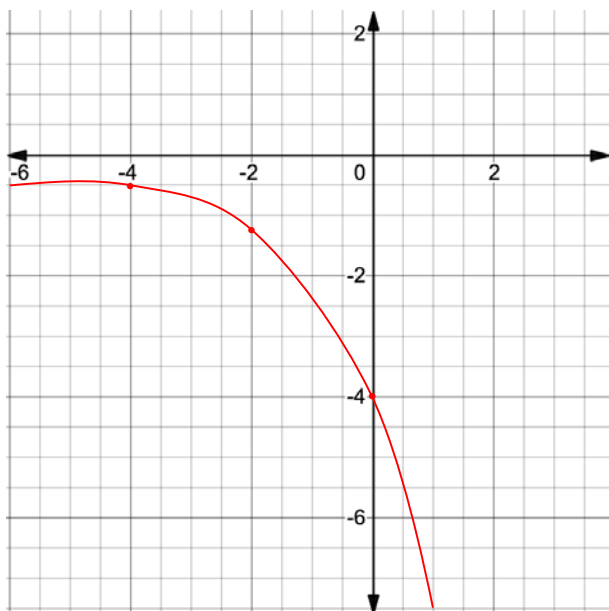


Algebra 1 Name **Answer Key** _____
 Unit 11 Date _____
 Lesson 3 Homework Period _____

For questions 1 – 2, consider the function $f(x) = -4(1 + 0.65)^x$.

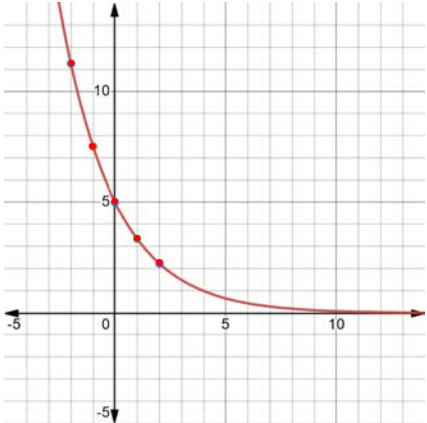
1. Since a is ☐ positive and b is ☐ between 0 and 1, the function
☒ negative ☒ greater than 1,
 is ☐ increasing. The left end behavior of the graph is ☒ $y \rightarrow 0$ and the
☒ decreasing. ☐ $y \rightarrow -\infty$
 right end behavior is ☐ $y \rightarrow 0$. The y -intercept of the function is ☒ -4 .
☒ $y \rightarrow -\infty$. ☐ 0.65 .
☐ 1.65 .

2. Sketch the graph of $f(x) = -4(1 + 0.65)^x$.



Graph the function of $y = 5\left(\frac{2}{3}\right)^x$, complete the table of values, and then determine all key features.

3. Graph



4. Table of Values

x	y
-2	11.25
-1	7.5
0	5
1	3 1/3 or 3.3
2	2 2/9 or 2.2

Commented [CO1]: <https://www.desmos.com/calculator/zgia8g7dv1>

Key Features	
5. Growth or Decay	decay
6. Domain and Range	$\mathbb{R}$ $y > 0$
7. Increasing/Decreasing	decreasing $(-\infty, \infty)$
8. Constant Percent Rate of Change	$\frac{2}{3}$ or $66\frac{2}{3}\%$
9. Left and Right End Behavior	as $x \rightarrow -\infty$, $y \rightarrow \infty$ as $x \rightarrow \infty$, $y \rightarrow 0$
10. Positive or Negative	positive only $(-\infty, \infty)$